

WORKING PAPER

Bounded by Design: Applying the Donut Model to Responsible AI Implementation in Enterprise Operations

A Practitioner Framework for AI Governance, Scale, and Accountability

Author	Azhar Shaikh
Affiliation	azee.ai Independent Researcher & AI Strategy Practitioner
Document Type	Working Paper / Practitioner Research
Year	2025
Keywords	Donut Model, AI governance, enterprise AI, responsible AI, operational excellence, bounded AI, transformation

Abstract

The adoption of artificial intelligence across enterprise operations has accelerated faster than the governance structures designed to contain it. Organizations deploy AI systems to drive efficiency, reduce headcount, and compress decision cycles — but rarely define the boundaries within which those systems should operate. This paper proposes a practitioner framework that adapts Kate Raworth's Donut Economics model to AI implementation in enterprise contexts, positioning organizational health as the social foundation and ethical-operational constraints as the ecological ceiling. The resulting Bounded AI Design (BAD) framework offers practical implementation guidelines for AI leaders seeking to deploy at scale without breaching the boundaries that sustain trust, regulatory standing, and workforce capability. The framework is illustrated through its application across three operational domains: shared services, process excellence, and workforce transformation.

1. Introduction

Enterprise AI adoption in 2025 sits at an uncomfortable juncture. The capability curve has outrun the governance curve. Organizations that rushed to deploy large language models, robotic process automation, and AI-assisted decision systems now find themselves managing a fragile architecture: systems that perform impressively in controlled conditions but introduce unforeseen risks when scaled, generalized, or left without meaningful human oversight.

The dominant discourse in AI implementation has been framed around enablement — what AI can do, how fast it can be deployed, and what cost efficiencies it can unlock. What is conspicuously absent from most enterprise AI strategies is an equally rigorous treatment of constraint: what AI should not do, where human judgment must remain sovereign, and how organizations can build systems that remain trustworthy under operational pressure.

This paper argues that this gap is not primarily a technical problem. It is a design philosophy problem. Enterprise AI strategies have been built on an expansion logic — more automation, more coverage, more speed — without an equivalent attention to the limits that make expansion sustainable. Kate Raworth's Donut Economics model, developed as a framework for sustainable human economies, offers a structurally analogous solution: a dual-boundary design that defines both a social foundation (below which systems must not fall) and an ecological ceiling (above which systems must not operate).

Adapted for enterprise AI, this produces what this paper terms the Bounded AI Design (BAD) framework. The name is deliberate. In practitioner contexts, 'bad AI' is often understood as AI that produces errors or harmful outputs. This paper reframes it: bad AI design is AI deployed without bounds. The framework operationalizes boundedness as a design principle rather than a compliance afterthought.

2. The Donut Model: A Brief Primer

Kate Raworth introduced the Doughnut Economics model in 2012, subsequently developed in her 2017 book of the same name. The model posits that human societies must navigate between two boundaries: a social foundation, representing the minimum conditions necessary for human wellbeing (access to food, education, healthcare, political voice), and an ecological ceiling, representing the planetary limits beyond which environmental destabilization becomes irreversible.

The 'donut' shape describes the safe and just space between these boundaries — the zone in which societies can thrive without shortfalling on human needs or overshooting natural systems. The model

has since been applied by cities (notably Amsterdam and Copenhagen), development organizations, and corporate sustainability teams as a planning and accountability tool.

The structural logic of the model translates directly to AI governance:

Just as economies must sustain a social floor without breaching an ecological ceiling, AI systems must operate above a capability floor that preserves meaningful organizational function while remaining below an autonomy ceiling that preserves accountability, ethics, and human primacy.

The translation requires redefining both boundaries for the enterprise context, which Section 4 addresses in detail.

3. The Problem with Current Enterprise AI Governance

Three structural failures characterize how most enterprises currently govern AI implementation.

3.1 Governance as Retrofit

The predominant pattern in enterprise AI adoption has been deployment first, governance later. Business units adopt AI tools for narrow use cases — contract review, customer triage, payroll processing — and governance teams are brought in after the fact to contain risks that have already materialized. This retrofit model creates fragmented oversight, inconsistent standards across functions, and governance frameworks that are perpetually behind the capability curve.

3.2 The Compliance Conflation

Many organizations treat AI governance as synonymous with regulatory compliance. This is a category error. Compliance sets a legal floor; governance sets an operational framework. An AI system can be fully compliant with data privacy regulations and still produce systematically biased hiring recommendations, erode workforce capability through over-reliance, or create accountability gaps when decisions are challenged. Compliance-as-governance leaves the most consequential risks unaddressed.

3.3 Absence of Organizational Health Metrics

Enterprise AI implementations are routinely assessed on efficiency metrics: cost reduction, throughput improvement, cycle time compression. Rarely are they assessed on organizational health

metrics: workforce capability over time, decision quality under failure conditions, employee trust in AI-assisted systems, or organizational resilience when AI tools are unavailable. The absence of these metrics means organizations are optimizing for narrow operational gains while systematically degrading the organizational conditions that make those gains sustainable.

These three failures share a common root: the absence of a dual-boundary design philosophy. Organizations know what they want AI to do. They have not adequately defined what AI must not do and what conditions must be preserved for the organization to remain viable.

4. The Bounded AI Design (BAD) Framework

The Bounded AI Design framework adapts Raworth's dual-boundary logic to enterprise AI governance. It defines two boundaries that any AI implementation strategy must respect:

- The Organizational Foundation — the minimum conditions of human capability, trust, and organizational function that AI systems must not erode
- The Autonomy Ceiling — the maximum scope of AI decision-making authority beyond which meaningful human accountability is lost

AI systems should operate in the space between these boundaries: capable enough to deliver operational value, constrained enough to preserve organizational integrity.

4.1 The Organizational Foundation (The Floor)

The organizational foundation defines the conditions that must be maintained for an enterprise to function as a capable, trustworthy institution over time. These conditions are not optional enhancements — they represent the minimum viable organizational health across six dimensions:

Dimension	Minimum Condition	Example Failure Mode
Human Capability	Core professional judgment must be exercised and developed, not replaced	Staff lose ability to perform tasks without AI assistance after 18 months
Workforce Trust	Employees must have sufficient transparency in AI systems affecting them	AI performance management system creates distrust, disengagement, attrition

Decision Accountability	Every AI-assisted decision must have an identifiable human accountable	Loan rejections processed by AI with no clear human owner for appeals
Institutional Knowledge	AI systems must not be sole custodians of critical organizational knowledge	System failure reveals no human understanding of the process it automated
Stakeholder Trust	External stakeholders must retain confidence in organizational integrity	Clients discover AI-generated advice was presented as human professional judgment
Regulatory Standing	AI deployment must not create latent legal exposure	AI hiring tools found to have produced discriminatory outcomes at scale

4.2 The Autonomy Ceiling (The Ceiling)

The autonomy ceiling defines the maximum scope of AI decision-making authority. It is not a fixed technical threshold — it is context-dependent, calibrated to the stakes of the decision, the quality of available oversight, and the organizational maturity of AI governance structures. The ceiling is defined across four dimensions of autonomy:

Autonomy Dimension	Below the Ceiling (Acceptable)	Above the Ceiling (Breach)
Decision Finality	AI recommends; human approves or overrides	AI decides; human notified after execution
Exception Handling	AI escalates anomalies to human judgment	AI resolves exceptions without human review
Scope Creep	AI operates within defined task parameters	AI extends scope to adjacent decisions without authorization
Self-Modification	AI system parameters changed by human governance	AI system adjusts its own thresholds based on output optimization
Stakeholder Impact	AI-assisted decisions reviewed before external effect	AI executes customer or employee-facing decisions without checkpoint

4.3 The Safe Operating Zone

The safe operating zone is the space between the organizational foundation and the autonomy ceiling. AI systems designed and governed within this zone can deliver genuine operational value without compromising the organizational conditions that make that value sustainable. The zone is not

static — it expands or contracts as organizational AI maturity develops, as governance structures strengthen, and as trust between human operators and AI systems accumulates through demonstrated reliability.

This dynamic quality is essential to the framework's practical utility. A newly deployed AI system in a nascent governance environment operates with a narrow safe zone — limited autonomy, strong human oversight, conservative exception handling. As the organization builds AI fluency, governance capability, and validated track records, the safe zone expands. This creates a principled pathway from AI augmentation to AI-assisted operations to, eventually, AI-led processes with robust human governance — rather than the current norm of undifferentiated deployment at whatever speed procurement allows.

5. Practical Implementation Guidelines

The following guidelines are organized across three phases of AI implementation: design, deployment, and sustained operation. They are intended as practitioner tools, not compliance checklists.

5.1 Phase 1: Design (Before Deployment)

5.1.1 Boundary Mapping

Before any AI system is deployed, the implementation team should conduct a boundary mapping exercise that identifies, for the specific use case, where the organizational foundation sits and where the autonomy ceiling should be set. This is not a risk assessment in the conventional sense — it is a design exercise that defines the operational parameters within which the AI system will be built.

Boundary mapping outputs a Bounded Design Brief (BDB) that specifies: the six organizational foundation conditions and the minimum acceptable state for each, the five autonomy dimensions and the maximum acceptable scope for each, and the escalation paths that activate when either boundary is approached.

5.1.2 Human-in-the-Loop Architecture

Every AI system deployed in the enterprise should have a documented human-in-the-loop architecture that specifies, at a decision-type level, the exact conditions under which human review is required. This document should be maintained alongside the technical architecture, reviewed

whenever system scope changes, and treated as a governance artifact, not merely an operational procedure.

5.1.3 Capability Preservation Design

For any process that AI is augmenting or replacing, the implementation design should specify how the human capability that the AI replaces will be preserved at a viable organizational level. This requires defining the minimum human competency that must be maintained, the frequency at which humans must exercise that competency independently, and the mechanisms by which organizational knowledge will be maintained outside the AI system.

5.2 Phase 2: Deployment

5.2.1 Staged Scope Expansion

AI systems should be deployed with a staged scope expansion protocol rather than a full-function launch. Each stage should have defined success criteria and boundary health checks before expansion proceeds. Specifically, organizations should verify that organizational foundation conditions remain above minimum thresholds and that no autonomy ceiling dimension has been breached before each expansion phase.

5.2.2 Boundary Health Monitoring

From day one of deployment, boundary health metrics should be tracked alongside operational performance metrics. A deployment that delivers a 30% throughput improvement while eroding workforce capability or reducing decision accountability is not a successful deployment — it is a boundary breach delivered with positive efficiency numbers. Standard dashboards should include both performance indicators and boundary health indicators.

5.2.3 Exception Handling Protocols

A comprehensive exception handling protocol should be established before deployment, not derived reactively. The protocol should address: what constitutes an anomaly that triggers human review, who is responsible for exception resolution, what the maximum acceptable response time is, and how exception patterns inform system improvement. Exceptions are the most common site of autonomy ceiling breaches — they are where AI systems, optimized for throughput, are most likely to extend beyond their sanctioned scope.

5.3 Phase 3: Sustained Operation

5.3.1 Annual Boundary Reviews

At minimum annually, and following any significant system change or incident, a formal boundary review should assess whether the organizational foundation conditions and autonomy ceiling settings remain appropriately calibrated. Organizational context changes: staff composition shifts, regulatory environments evolve, AI system capabilities improve. Boundary settings that were appropriate at deployment may be too conservative or too permissive twelve months later.

5.3.2 Workforce Capability Audits

The most persistently undertracked dimension of enterprise AI deployment is its effect on human capability over time. Organizations should conduct periodic capability audits that assess, specifically, whether staff working with AI systems have maintained the ability to perform their core functions without AI assistance. These audits should be structured as operational tests, not self-assessment surveys. The findings should feed directly into capability preservation programming.

5.3.3 AI Ethics Review Cadence

AI systems operating in high-stakes domains — decisions affecting employees, customers, or regulatory standing — should be subject to a structured ethics review cadence that examines output patterns for systematic bias, unintended scope extension, and stakeholder harm. This review should be conducted by a team that includes operational leaders, not solely technology or compliance professionals. The practitioner perspective is essential to identifying boundary breaches that technical reviews routinely miss.

6. Application Across Three Operational Domains

The following section illustrates the BAD framework's application across three enterprise operational domains relevant to shared services, process excellence, and workforce transformation contexts.

6.1 Shared Services Operations

In shared services environments, AI deployment typically targets transaction processing, query resolution, and exception management. The organizational foundation concern most salient here is institutional knowledge: shared services functions are knowledge-intensive, and AI systems that absorb process logic without human custodianship create brittle operations. The autonomy ceiling concern is decision finality: in high-volume transaction environments, the efficiency pressure to remove human review from final decisions is strongest, and the risks of doing so are most consequential at scale.

The BAD framework applied to shared services specifies: at minimum, one trained human practitioner per process family who can operate the process without AI assistance; a hard decision review requirement for any transaction above a defined financial or regulatory risk threshold; and a quarterly knowledge validation exercise that confirms AI system parameters are understood and owned by named human practitioners.

6.2 Process Excellence Functions

In process excellence functions, AI deployment typically targets process mining, performance analytics, and improvement opportunity identification. The organizational foundation concern most salient here is human capability: process excellence practitioners who rely entirely on AI-generated diagnostics progressively lose the analytical instinct that makes them effective. The autonomy ceiling concern is scope creep: AI process mining tools, once deployed, tend to be extended into recommendation and then into implementation support, often without the governance checkpoints that these scope extensions warrant.

The BAD framework applied to process excellence specifies: AI tools should be used for data surfacing and pattern identification, with human practitioners retaining analytical synthesis and recommendation responsibilities; any extension of AI tool scope from diagnostic to advisory to implementation requires a formal boundary review; and practitioner development programs should maintain manual process analysis skills as a core competency alongside AI tool literacy.

6.3 Workforce Transformation

In workforce transformation contexts, AI deployment typically involves workforce analytics, skills gap identification, talent matching, and performance management. This is the highest-stakes domain for boundary management because the decisions involved directly affect individuals' livelihoods and careers. The organizational foundation concern most salient here is decision accountability: every AI-assisted people decision must have a human accountable for it who understands the basis on which the recommendation was made. The autonomy ceiling concern is the most fundamental: AI systems should never make final determinations about individuals — hiring, promotion, performance rating, or termination — without meaningful human review.

The BAD framework applied to workforce transformation specifies: a mandatory human review step for every AI-assisted people decision, with documentation of the human's independent assessment; prohibitions on AI-generated performance scores being used as the sole or primary basis for employment decisions; and regular output audits that examine AI recommendations for patterns that may reflect systemic bias in training data or optimization criteria.

7. Limitations and Future Directions

The BAD framework is a practitioner framework, not a formally validated research instrument. Its boundary conditions are illustrative rather than empirically derived, and the specific minimum thresholds for organizational foundation dimensions will vary significantly by industry, organizational size, regulatory context, and AI system type. Organizations adopting the framework should treat it as a design philosophy and governance structure to be calibrated to their specific context, not as a prescriptive standard.

Three areas warrant further development. First, quantitative operationalization of the boundary health metrics proposed in Section 5 requires empirical work across organizations that have deployed the framework. Second, the dynamic calibration of the safe operating zone — the mechanism by which organizational AI maturity translates into appropriate boundary adjustments — requires a more developed maturity model than is proposed here. Third, the intersection of the BAD framework with emerging regulatory requirements, notably the EU AI Act and evolving ASEAN AI governance frameworks, requires detailed mapping that this paper does not undertake.

The author intends this paper as a starting point for practitioner dialogue, not a finished theoretical contribution. Feedback, critique, and empirical application are invited through azee.ai.

8. Conclusion

Enterprise AI strategy has been dominated by an expansion logic for long enough that its costs are now visible. Organizations that deployed AI to capture efficiency gains are encountering the boundary effects those deployments were never designed to manage: eroded human capability, diffuse accountability, workforce distrust, and governance frameworks that are perpetually reactive.

The Donut Model offers a corrective design philosophy. Its dual-boundary logic — sustain a foundation, respect a ceiling, operate in the safe space between — is not a constraint on AI's potential. It is the structure that makes AI's potential organizational and durable. An AI system that operates below the organizational foundation is doing organizational harm beneath the threshold of visibility. An AI system that operates above the autonomy ceiling is doing governance harm beyond the reach of oversight.

Bounded by design, AI systems can do what the expansion logic always promised but rarely delivered: genuine, sustained, accountable improvement in how organizations work. The boundary is not the obstacle. It is the condition of possibility.

References and Further Reading

Raworth, K. (2017). Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist. Chelsea Green Publishing.

European Commission. (2024). EU Artificial Intelligence Act. Official Journal of the European Union.

OECD. (2023). OECD Principles on Artificial Intelligence. Organisation for Economic Co-operation and Development.

Brynjolfsson, E., & McAfee, A. (2014). The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies. W.W. Norton & Company.

Wachter, S., Mittelstadt, B., & Russell, C. (2017). Counterfactual Explanations Without Opening the Black Box: Automated Decisions and the GDPR. Harvard Journal of Law & Technology, 31(2).

Zuboff, S. (2019). The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power. PublicAffairs.

Accenture / World Economic Forum. (2023). Responsible AI Governance Principles for Enterprises. World Economic Forum White Paper.

About the Author

Azhar “Azee” shaikh is a senior transformation and operational excellence practitioner with over 20 years of experience across Asia Pacific, including enterprise-scale AI and process transformation programs in shared services, financial services, and BPO environments. He holds an MBA from IIM Bangalore, is completing doctoral research, and holds credentials including CAIDO and membership

of the HBR Advisory Council. He publishes on AI strategy, governance, and enterprise transformation. He has worked in large insurance company(QBE as SVP for transformation) and Cognizant and HP in past..

Contact: me@azee.ai | Published as Working Paper 2025